Grant Proposal: Private Lending on Aleo

PWN DAO Foundation · Aleo SCALE grant programme · 2 November 2026

Grant request	Delivery period	Main deliverables
$85,000	17 weeks from project start	Aleo programs, TypeScript SDK, reference app, external audit and mainnet launch

Project Overview

We are building private peer-to-peer lending infrastructure for Aleo. The protocol connects asset holders seeking liquidity with lenders offering fixed-rate credit, starting with ALEO collateral and USDCx loans. Both lenders and borrowers can post offers. Each offer can be public or private and can fund multiple independent loans.

How it works. A lender escrows credit or a borrower escrows collateral when posting a funded offer. The accepting counterparty supplies the other asset. Each fill creates a loan with its own principal, collateral, fixed APR and duration. Repayment returns collateral to the borrower; an unpaid loan can be settled through a lender claim at or after maturity. There are no price oracles or price-triggered liquidations.

Privacy. Aleo's private records and authorization reduce direct public links between wallets and loan positions. Submitted loan economics and necessary escrow metadata remain public, and timing or amount correlation is possible. The Privacy Model details these boundaries.

Grant scope. The grant takes the working proof of concept through production rework, live USDCx integration, proof-enabled public-network validation, an independent security audit and mainnet launch. Deliverables include Leo programs, a TypeScript SDK for wallet and application integrations, and a reference app. All software will be released under GPL-3.0. The delivery plan includes test evidence and external user testing.

Why it matters. Holders can obtain liquidity without selling their assets, while lenders can earn an agreed return over a fixed term. The protocol brings this model to Aleo with less direct public exposure of wallet relationships. Pilot lenders, borrowers and completed loans will test demand for the service.

Team Background

PWN DAO Foundation has built crypto lending infrastructure for over 5 years. Its peer-to-peer lending protocol, PWN, is deployed on 13 EVM mainnets. It has processed more than $10m in loan volume for over 5,000 users. The same team builds OWN, crypto-backed mortgage infrastructure running on PWN's lending primitives.
Josef Jelacic: Founder of PWN DAO and OWN; previously at the Ethereum Foundation. Project role: protocol architecture and design oversight. LinkedIn.
Šimon Kozák: Founding Engineer and CTO at PWN DAO and OWN. Project role: protocol and application development. LinkedIn · GitHub.
Jan Mikšík: Engineer with four years of experience building DeFi lending applications. Project role: protocol and application development. LinkedIn · GitHub.

Market Potential & Demand

Why now

Private stablecoins are live on Aleo: USDCx is backed by USDC in Circle xReserve, and USAD is issued by Paxos Labs and backed by USDG. Their combined on-chain supply was approximately **$4.71m at peg on 2 November 2026**. USDCx state · USAD state.
Trading and liquid-staking products provide adjacent infrastructure: potential exit liquidity for collateral and staking assets that borrowers could use without giving up reward exposure.

Target users and use cases

1. **Long-term ALEO holders and stakers.** Borrow USDCx against ALEO without selling. Planned pALEO and stALEO integrations would extend this use to holders retaining exposure to staking rewards. Native staking positions are a later roadmap item.
2. **Stablecoin holders and treasuries.** Allocate funds to collateralized loans at an agreed rate and term, choosing offers that match their liquidity needs and risk tolerance. This could serve holders managing payroll, settlement or treasury balances.
3. **Private bilateral credit.** Funds, market makers, DAOs and OTC desks can arrange credit with a chosen counterparty through private offers, whether they are seeking funds or offering them.
4. **Holders of bridged assets.** Planned collateral integrations could let holders of ETH, wBTC and SOL on Aleo borrow without selling.
5. **Applications that embed credit.** Wallets and treasury applications can integrate collateralized borrowing and lending through the SDK without building their own lending protocol.

6. **Regulated lenders.** Institutions requiring transfer auditability can explore integration with USDCx's compliance-observer model while reducing public exposure of wallet relationships.

Collateral and credit assets

Asset	Availability (2 Nov 2026)	Example Role
ALEO (native)	Native ALEO through credits.aleo	First collateral asset
USDCx / USAD	Live private stablecoins backed by USDC and USDG, respectively	USDCx: initial credit asset; USAD: planned follow-on
pALEO (Pondo)	Pondo liquid-staking token	Planned follow-on collateral
stALEO (Beta Staking)	Beta Staking liquid-staking token	Planned follow-on collateral
ETH, wBTC, SOL on Aleo	Prospective bridged-token integrations; swap ecosystem	Planned follow-on collateral

Market context

- **Aleo:** Native ALEO and planned liquid-staking integrations form the prospective collateral base; existing stablecoins provide dollar-denominated credit assets.
- **Web3:** Galaxy Research reported $56.16bn in outstanding crypto-collateralized loans at the end of Q2 2026, including $20.43bn in DeFi applications. The total spans CeFi, DeFi and crypto-backed CDP stablecoins.
- **Web2:** Borrowing against investment portfolios is an established private-banking use case. Federal Reserve researchers estimated $138bn in US securities-based loans in Q1 2024, illustrating demand for liquidity without selling portfolio assets.

These supply and adjacent-market figures describe different populations and lending arrangements. They provide context, not a combined addressable market or evidence of demand for this protocol. Pilot commitments and completed loans will establish demand.

Ecosystem Alignment

Impact

- **Lending infrastructure.** Fixed-term credit expands the uses of private dollars, ALEO and compatible staking assets. It supports SCALE's Drive

Commercialization and Adoption and Invest in Future Infrastructure (ZeFi) objectives and its DeFi scope.

- **Payments and treasury.** Collateralized credit and lending give payment and treasury applications an additional use for on-chain balances.
- **Open infrastructure.** The Leo programs, SDK and module interface will be open source. Teams can add offer modules, embed lending through the SDK or deploy their own markets. Published privacy and security findings will support other Aleo developers.

Readiness

- **No oracle dependency.** Loans settle on time and repayment, not price, so launch does not wait for price-feed infrastructure.
- **Built on existing standards:** `credits.aleo`, the token registry, and Provable's compliant stablecoin programs behind USDCx and USAD.
- **Working PoC.** The historical testnet deployment and current local multi-token implementation provide a starting point for production work. Demonstrated capabilities and validation limits are listed in Current Implementation and Evidence.

Collaborative Value

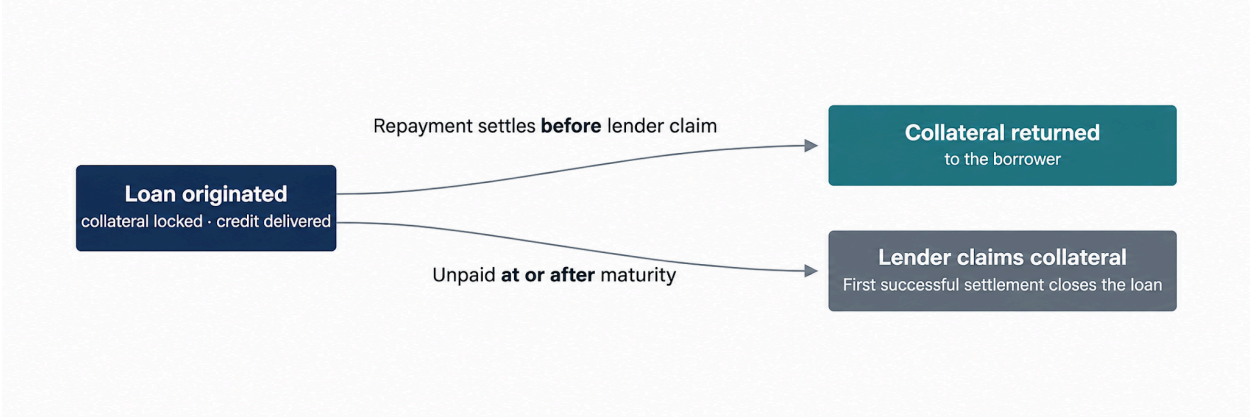
Potential integration targets	**Planned integration**
Circle (USDCx), Paxos (USAD)	USDCx as the initial credit asset; USAD as a follow-on integration
Provable: Shield wallet, Shield Swap, delegated proving	Reference-app wallet support, potential collateral exit liquidity and delegated proving
Pondo, Beta Staking	pALEO and stALEO as collateral, giving liquid-staking tokens new utility
Request Finance, Toku, Zebec, Bitwave, Utila	Treasury yield and embedded credit through the SDK
zPass, ANS (future)	Identity-gated private offers and readable counterparty names

Potential integration targets	Planned integration
Everstake and validators (future)	Native staking positions as collateral

System Design & Specification

Loan lifecycle and funded offers

Each loan fixes its credit and collateral assets, principal, collateral requirement, APR, duration and repayment rule at origination. Repayment returns collateral to the borrower and allocates repayment to that loan's lender. Repayment remains possible before or after maturity until a lender claim settles. The lender may claim collateral at or after maturity; the first successful settlement prevents the other. Maturity alone does not transfer collateral.



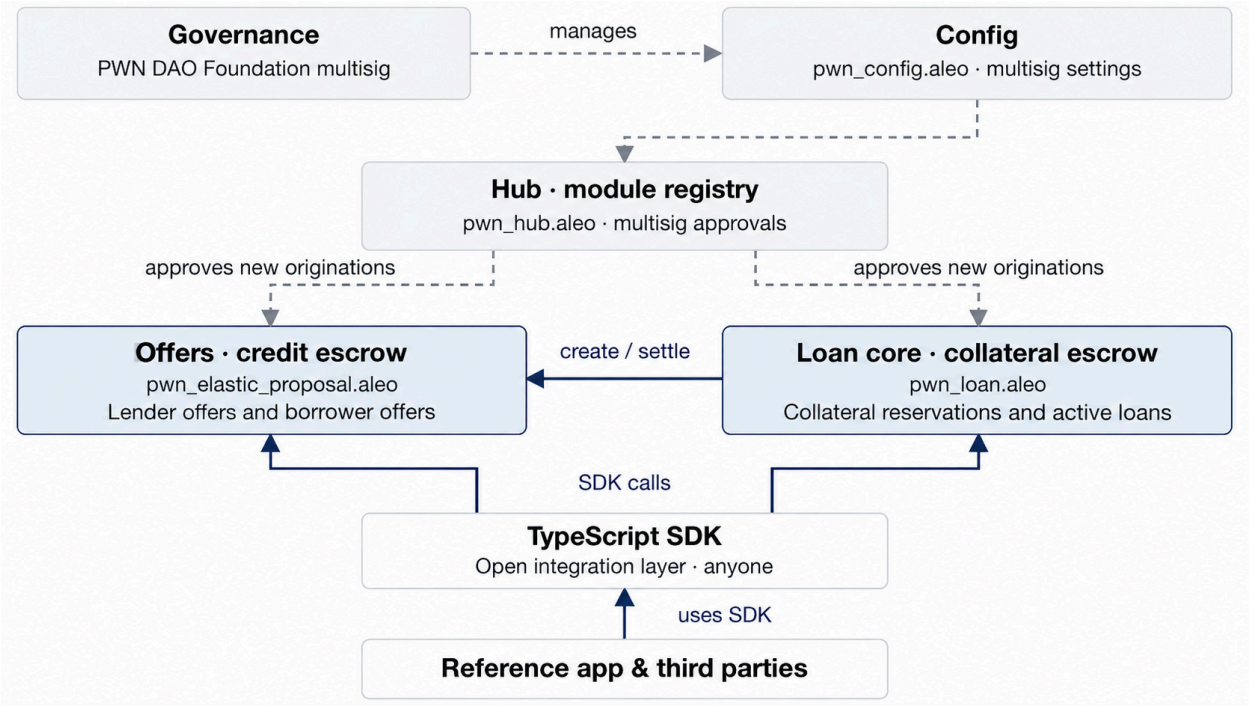
The funded baseline follows the Elastic Proposal model. A lender offer escrows credit and specifies fill limits, an asset pair, collateral ratio, duration and expiry. A borrower offer instead escrows collateral and proposes borrowing terms to lenders, with its own fill limits and expiry. The taker supplies the complementary asset at acceptance. Both directions allow repeated fills, each creating an independent loan with separate collateral, maturity and lender repayment rights. Borrower-offer repayments belong to each matched lender, not the offer maker. Expiry stops new fills; cancellation refunds only unused escrow and leaves active loans intact. Capacity is non-revolving: repayment and returned collateral do not replenish it.

Lenders and borrowers agree the collateral ratio and terms. Lenders bear collateral-price risk in exchange for the agreed return; price moves cannot trigger liquidation. The parties choose the assets and collateralization they accept without ongoing price-feed updates.

Visibility is independent of offer direction. Public offers can serve different takers. Private offers use the same funded model in either direction. Private authorization restricts acceptance to one designated counterparty, who can fill repeatedly. Their payloads are shared off-chain rather than listed publicly.

Optional funding at acceptance. A local public-USDCx allowance prototype demonstrated atomic funding and private native-collateral locking without prefunding principal or collateral; transaction fees apply. Lender address and amount are public; balances remain unreserved. Acceptance needs borrower authorization and rolls back on revoked allowance or insufficient funds. M1 assesses wallet, live-issuer and public-network feasibility. Technical evidence.

Modules, custody and on-chain data access



Layer	Responsibility	Authority boundary
Hub and Config	Module authorization, zero protocol fee and a stop for new originations.	Foundation multisig manages module authorization; administration cannot block or redirect settlement of existing positions.
Proposal program	Holds lender credit escrow, enforces fill limits, refunds unused credit and allocates per-loan repayments; coordinates borrower reservations with the loan core.	The maker controls unused escrow; each matched lender owns its loan's repayment rights. Cancellation cannot reclaim active-loan collateral.

Layer	Responsibility	Authority boundary
Loan core	Holds pending borrower collateral and active loan collateral; enforces repayment, maturity and one-time settlement.	Loan terms are fixed at origination.
Token programs	Native and private token transfers and program-held public balances.	Transfer execution and issuer controls are governed by each token program.
SDK and applications	Wallet connection, proof and record handling, public discovery, private-offer delivery/import, recovery and user journeys.	Public state determines settlement; required off-chain data is described below.

Escrow uses program-controlled public balances; withdrawals deliver private records to users. Programs never hold spendable private records. Offer types are separate proposal modules authorized through the Hub, allowing new types without changes to the loan core. The production core will be audited before immutable deployment.

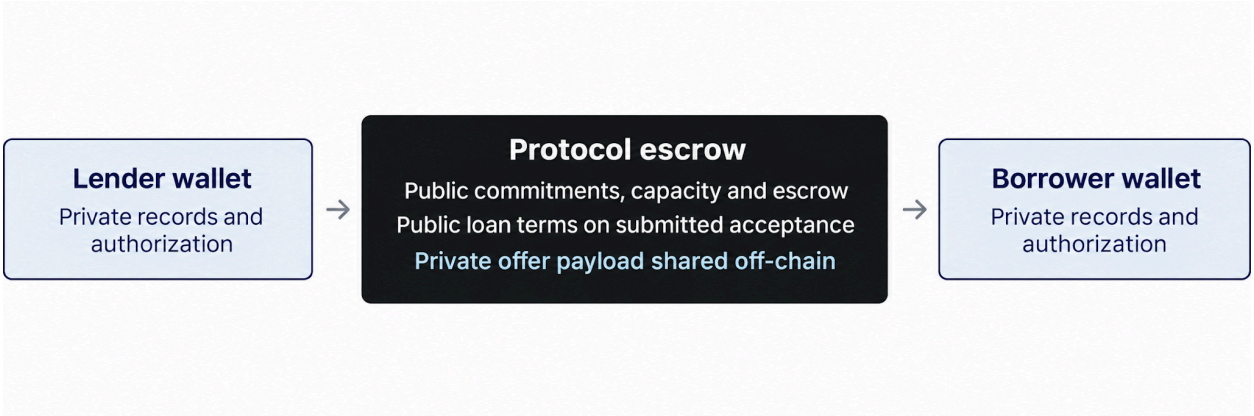
The SDK will read public offers, capacity, loan terms and status from Aleo state for the reference app and third-party integrations. An optional indexer can accelerate public search, but the SDK verifies terms and status against chain state before preparing acceptance. Private-offer payloads must match their on-chain commitments. Public borrower offers also need a shared opening containing payout details; once shared, the maker can be offline during acceptance. M2 covers encrypted delivery and import, private-record backup, and recovery of per-position authorization data for lenders and borrowers.

Asset scope

The initial production pair is ALEO collateral against USDCx credit. The PoC covers native ALEO, ARC-20 and USDCx/USAD token profiles using pinned local fixtures. Planned integrations include USAD credit and pALEO, stALEO and bridged-token collateral. There is no administrative asset allowlist. Compatibility requires exact supported token descriptors, isolated accounting and exact transfer-delta checks. Integrations must assess liquidity, bridge/issuer risks and, for staking tokens, reserves and withdrawals. Issuer freeze, burn, pause and upgrade powers remain risks; M1 defines the token-deficit policy and validates live issuer behavior before deposits. NFTs, rebasing and fee-on-transfer tokens are outside V1.

Privacy model

V1 uses Aleo records and private authorization to reduce direct public links between wallet addresses and loan positions. The table describes funded offers; the optional public-allowance prototype also exposes its lender address and funding amount.



Data	Visibility in V1
Participant addresses	Private inputs, records and salted authorization avoid direct wallet addresses in public lending fields. Counterparties receive necessary payout information. Public fee payments can reveal the payer.
Balances and position ownership	User-held records and authorization data are private; program escrow balances are public.
Public/private offers and submitted loan economics	Public offers advertise terms. Private payloads are shared off-chain; commitments, expiry, status, necessary capacity and escrow asset/amount may remain visible. Submitted loan economics are public, including rejected acceptances.
Issuer and prover visibility	The issuer’s designated compliance observer receives transfer data. Delegated provers see the inputs entrusted to them, so prover choice is a separate boundary from what appears on-chain.
Timing and amount correlation	Public escrow, amounts, timing and counterparty knowledge can reveal links between positions. Private records alone do not guarantee unlinkability.

Privacy references: PoC disclosure matrix; Aleo records and scoped view keys; delegated proving; stablecoin compliance visibility.

Current implementation and evidence

Implementation	Demonstrated scope	Next step
Public-testnet predecessor	Four single-pair programs using native ALEO credit and live token-registry collateral; funding, draw, repayment, default claim, refund and withdrawal executed with real proofs.	Public-testnet validation of the multi-token design and live USDCx lifecycle.
Current multi-token PoC	Local tests across native ALEO, ARC-20 and USDCx/USAD profiles cover both offer directions, repeated fills, accounting, adversarial and arithmetic cases. Selected executions generate real proofs; the devnode does not verify proofs, deployment certificates or consensus.	Proof-enabled public-network tests, wallet integration and live compliance validation.
Production target	Grant deliverable: audited immutable core, lender and borrower offers in public and private modes, production SDK and reference app.	Replace the PoC's timelocked upgrades with immutable deployment after audit.

Roadmap beyond V1

Future modules can add installments, refinancing, crowdfunded loans, borrower and lender hooks, and identity-gated private offers using zPass. A dedicated native-staking collateral module would handle reward accounting and unbonding.
Design references: PWN protocol architecture; PWN Elastic Proposal semantics; Provable compliant stablecoin programs; PoC interface decisions; PoC security status.

Funding & Milestones

The request is **$85,000 over 17 weeks** from the agreed project start. It funds a full review and production rework of the PoC, completion of the lending features, live token integrations, the

SDK, reference application, testing and independent security audit. Milestones are verified through published code, test results and deployment evidence.

Milestone	Target completion	Amount	Deliverable
M1 · Core and interfaces	Week 7	$35,000	Production candidate, both offer directions and visibility modes, live USDCx validation, tests and threat model
M2 · SDK, offer discovery and recovery	Week 10	$15,000	TypeScript SDK, public discovery, encrypted private-offer delivery/import and position recovery
M3 · Reference application	Week 13	$15,000	Lender and borrower journeys, selective disclosure and external testing
M4 · Audit and launch	Week 17	$20,000	External audit, remediation, retesting, immutable deployment and end-to-end mainnet loan
Total	17 weeks	$85,000	

M1 · Core protocol completion and interface freeze

Scope. Review all PoC programs and rework their architecture, interfaces and implementation for production: both offer directions and visibility modes, Hub/Config behavior, custody and settlement interfaces, and live USDCx integration. Retain module authorization and the token safeguards in Asset Scope, without administrative asset admission. Define issuer-risk and token-deficit policies; publish invariants, a threat model and the disclosure specification. Add proof-enabled public-testnet, adversarial, property and fuzz tests. Assess optional funding-at-acceptance feasibility and moving public witness fields into private inputs bound to blinded commitments.

Acceptance. Public-testnet evidence covers both offer directions: multiple public takers, repeated private fills and rejection of unauthorized acceptance; independent loan collateral, maturity and matched-lender repayment rights; non-revolving capacity, expiry, unused-escrow cancellation and overdraw protection; settlement timing, including rejection of premature claims, repayment before/after maturity until claim settlement, and the rule that the first successful settlement wins; double settlement, replay, administration changes and token-specific failures.

Demonstrate a live USDCx loan from funding through repayment with compliance outputs documented. Publish the privacy review, including rejected transactions, witness-privacy assessment and allowance feasibility with wallet/live-issuer constraints; freeze the core for audit.

M2 · SDK, offer discovery and recovery

Scope. A TypeScript SDK for publishing and accepting both offer directions, public discovery, capacity and loan-state queries, funding, repayment, lender claims, cancellation and withdrawals. Include encrypted private-offer delivery/import, commitment verification, fresh salts per commitment and loan, public-borrower opening/payout handling, private records and fees, USDCx proof preparation, and user-controlled backup/recovery of position authorization data.
Acceptance. Publish the SDK and integration documentation. Run the full testnet lifecycle without a proprietary backend, using shared payloads where required. Verify private-offer commitments and public-borrower acceptance while its maker is offline. Demonstrate privacy guards for successful and rejected calls, private fees on every lending path with no silent public fallback, backup recovery, and actionable errors for stale compliance data, unavailable funds and failed transactions.

M3 · Reference application

Scope. Wallet connection, lender and borrower offers, public browsing, encrypted private-offer sharing and acceptance, fills, repayment, lender claims, position overview and selective disclosure to a chosen party. Provide guidance on avoidable privacy leakage.
Acceptance. External testers complete lender and borrower journeys on testnet without team assistance, including public and private offers and repeated fills. Publish the app and usage documentation, fix blocking usability findings, and demonstrate single-loan disclosure without an account-wide view key. Agree tester count, supported wallets and usability criteria with Aleo before M3 begins.

M4 · Security review and mainnet launch

Scope. Independent audit of the production core and its integration boundaries, remediation and retesting, non-upgradeable mainnet deployment, and SDK package release. Audit preparation starts after the M1 interface freeze.
Acceptance. Publish the audit and remediation status, with material findings resolved before mainnet deposits. Publish program IDs, source and build references, and constructor policies. At least one real mainnet loan with external participants originated, repaid and settled end to end. Publish integration and operational documentation.

Success metrics

Targets for the six months after mainnet launch:

- 50 loans originated
- $250k in cumulative loan principal, measured in USD across supported assets
- 2 third-party apps integrating the SDK